SKILL 1 2400030484 - P. JIJNESH

In this experiment, I learned how to use **Git Version Control** to manage and track changes in project files. Git helps developers maintain different versions of their code and collaborate efficiently without losing previous work. The main objective of this experiment was to understand how to initialize a Git repository, add files, commit changes, create branches, merge them, and push the project to GitHub.

First, I created a new project folder and initialized it as a Git repository using the git init command. Then I configured Git with my username and email so that all commits are properly identified. After setting up the repository, I created two files named **Queues.java** and **doc.txt**, and added sample content to them.

Next, I checked the repository status using the git status command, which showed the newly created files as untracked. These files were then added to the staging area using the git add command and committed to the repository using the git commit command with an appropriate commit message.

After committing the files locally, I created a new repository on GitHub and connected my local repository to the remote repository using the git remote add origin command. The project files were then uploaded to GitHub using the git push command.

Further, I created a new branch called **feature-update** and made modifications to the project files. The changes were staged and committed. Similarly, another branch called **bug-fix** was created to simulate fixing an issue in the project. After making changes, both branches were merged back into the **main** branch using the git merge command.

Finally, I verified the commit history and ensured that all changes were successfully merged and stored in the Git repository.

Commands Used in the Project

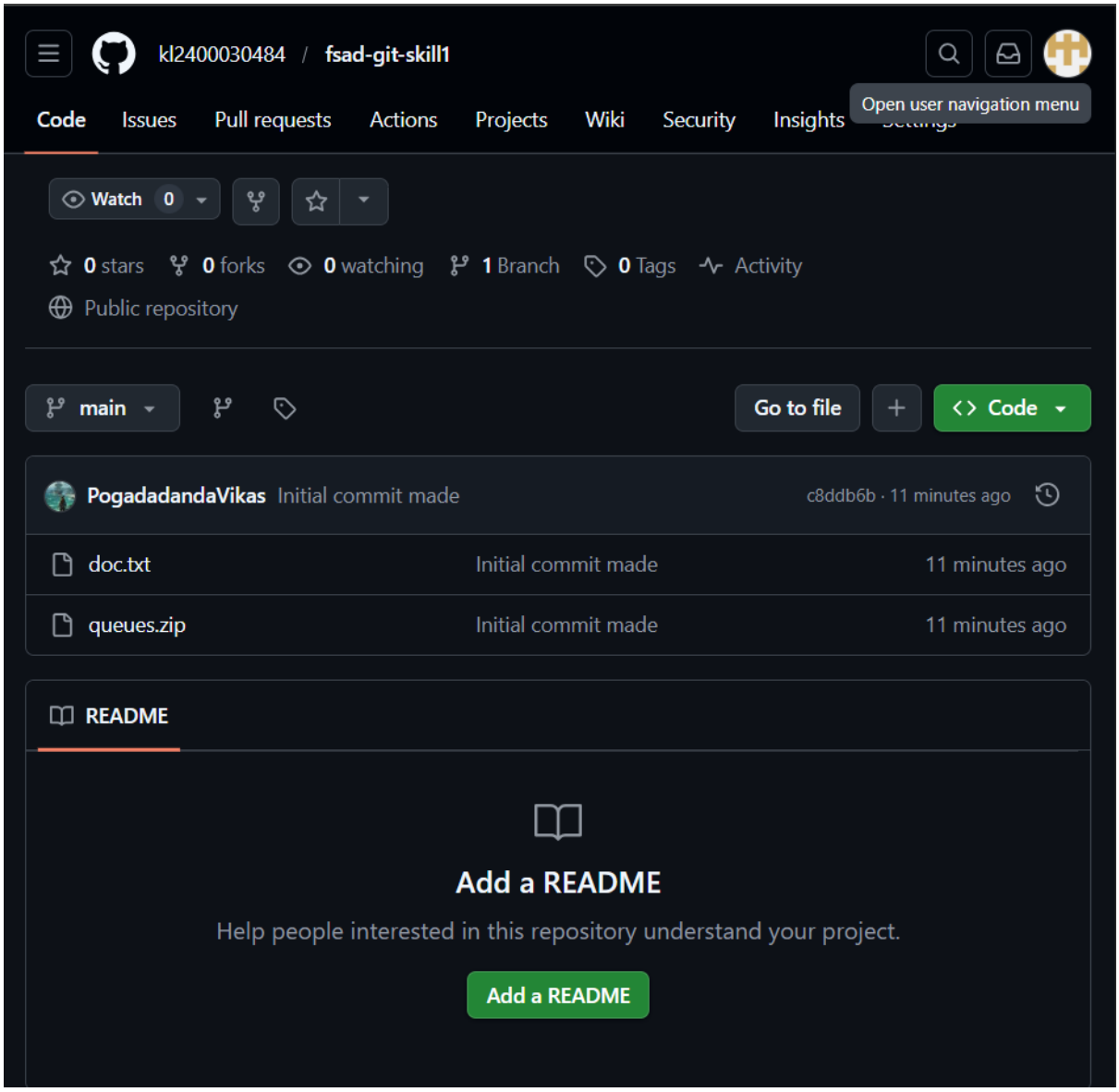
```
mkdir fsad-git-skill1
cd fsad-git-skill1
git config --list
git status
git add .
git commit -m "Initial commit "

git remote add origin https://github.com/kl2400030484/ fsad-git-skill1
git branch -M main
git push -u origin main

git checkout -b feature-update
echo "Feature update added"
git commit -m "Added feature update"
git checkout main
git checkout -b bug-fix
echo "Bug fixed here"
git add .
git commit -m "Bug fix applied"
git checkout main
git merge feature-update
git log
```

Screenshots

The below screenshots show the step-by-step process of performing the experiment, including repository creation, file creation, staging, committing, branch creation, merging, and pushing the code to GitHub.



C:\Windows\System32\cmd.e

```
Microsoft Windows [Version 10.0.26200.7922]
(c) Microsoft Corporation. All rights reserved.

D:\2400030484-github\skillprojects\fsad-git-skill1>git init
Initialized empty Git repository in D:/2400030484-github/skillprojects/fsad-git-skill1/.git/

D:\2400030484-github\skillprojects\fsad-git-skill1>git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

D:\2400030484-github\skillprojects\fsad-git-skill1>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=vikaspogadadanda@gmail.com
user.name=Vikas
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true

D:\2400030484-github\skillprojects\fsad-git-skill1>touch main.java notes.txt
'touch' is not recognized as an internal or external command,
operable program or batch file.

D:\2400030484-github\skillprojects\fsad-git-skill1>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        doc.txt
        queues.zip

nothing added to commit but untracked files present (use "git add" to track)

D:\2400030484-github\skillprojects\fsad-git-skill1>git add .

D:\2400030484-github\skillprojects\fsad-git-skill1>git status
On branch master

No commits yet
```

C:\Windows\System32\cmd.e

```
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   doc.txt
        new file:   queues.zip


D:\2400030484-github\skillprojects\fsad-git-skill1>git commit -m "Initial commit made"
[master (root-commit) c8ddb6b] Initial commit made
 2 files changed, 1 insertion(+)
 create mode 100644 doc.txt
 create mode 100644 queues.zip

D:\2400030484-github\skillprojects\fsad-git-skill1>git remote add origin https://github.com/kl2400030484/fsad-git-skill1.git

D:\2400030484-github\skillprojects\fsad-git-skill1>git branch -M main

D:\2400030484-github\skillprojects\fsad-git-skill1>git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 2.48 KiB | 362.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kl2400030484/fsad-git-skill1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

D:\2400030484-github\skillprojects\fsad-git-skill1>git checkout -b feature-update
Switched to a new branch 'feature-update'

D:\2400030484-github\skillprojects\fsad-git-skill1>echo "feature updated"
"feature updated"

D:\2400030484-github\skillprojects\fsad-git-skill1>git commit -m"added feature update"
On branch feature-update
nothing to commit, working tree clean

D:\2400030484-github\skillprojects\fsad-git-skill1>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\2400030484-github\skillprojects\fsad-git-skill1>git checkout -b bug-fix
Switched to a new branch 'bug-fix'

D:\2400030484-github\skillprojects\fsad-git-skill1>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\2400030484-github\skillprojects\fsad-git-skill1>git merge feature-update
Already up to date.

D:\2400030484-github\skillprojects\fsad-git-skill1>git log
commit c8ddb6b335b16c2397cdc60dba4faa02a4b6c4b0 (HEAD -> main, origin/main, feature-update, bug-fix)
Author: Vikas <vikaspogadadanda@gmail.com>
Date:   Thu Mar 5 14:23:53 2026 +0530

    Initial commit made

D:\2400030484-github\skillprojects\fsad-git-skill1>
```